Bounded Closure of the $\theta_c = \ln \phi$ Lock: A Total-Effort Survey and a Structural-Kinship Result with Scale-Factor Self-Dual Cosmology

LUMENPIXEL
lumenpixel@proton.me / github.com/LUMENPIXEL-001
Independent researcher, Busan, Republic of Korea

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Abstract

This paper is a spin-off note in the LVC working-paper series, parallel to the dark-matter and the closure-as-asymptote spin-offs. It is not part of the LVC main line (f-v1.1, f-v1.2). The paper records the total-effort attempt to derive $\theta_c = \ln \phi$, the empirically preferred modulation centre of the LVC construction, from internal principles of the closed-universe theory. Two results are recorded. First, a single algebraic axiom $z = a(z)$ at the modulation centre — frame matching between redshift and scale factor — admits the unique positive solution $z_c = 1/\phi$, equivalently $\theta_c = \ln \phi$, without imposing the golden ratio as an input; the axiom carries all v1.1 modulation locks $(\theta_c, A, w, \Omega_m)$ as algebraic consequences once the v1.1 width relation $w = A/\ln \phi$ is invoked, and admits eight equivalent algebraic forms verified to machine precision. Second, the frame-matching condition is compared in detail to the scale-factor self-duality condition of Camara–Lima–Sotkov (CLS) cosmology. Direct numerical analysis on the ΛCDM background shows that the two conditions identify the same redshift only at $\Omega_m \approx 0.654$, in disagreement with Planck and with the LVC-locked value at the $\sim 50\sigma$ level. The two are therefore structurally analogous (both are fixed-point constructions on a duality-like map acting on cosmological variables) but not mathematically equivalent on the standard cosmological background. The status of the $\theta_c = \ln \phi$ lock after this total-effort attempt is recorded as: closed at the level of a single axiom internal to the closed-universe theory; not closed by embedding into the CLS framework; and not closed in the strong sense of a forcing derivation from a deeper principle. A closing remark records the structural context in which this status appears unsurprising.

1 Introduction

The localised late-time expansion-rate modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ documented across the v1–v16 phenomenological line of the LVC series, and the exact topological reconstruction in v1.1 [1], share one imposed input: the modulation centre at $\theta_c = \ln \phi$ in the logarithmic-redshift coordinate $\theta = \ln(1 + z)$, with $\phi = (1 + \sqrt{5})/2$ the golden ratio. The v1.1 paper records this value as an empirical lock, not as a derivation from the action. The closure note [3] records, separately, the categorical structure that would in general obstruct such a derivation in any single-universe theory.

The present paper is a spin-off documenting a total-effort attempt to derive $\theta_c = \ln \phi$ from internal principles. The work surveyed candidate mechanisms across 35-plus distinct constructions, traced eight equivalent algebraic forms of a single axiom that does succeed in closing the lock at the algebraic level, and compared the resulting axiom against the scale-factor self-duality framework of Camara, Lima and Sotkov (CLS) [4, 5, 6], in which a structurally similar fixed-point condition appears as the defining symmetry of a class of cosmological models.

The paper records two facts. The algebraic axiom $z = a(z)$ at the modulation centre — frame matching between the redshift coordinate and the scale-factor function — admits the unique positive solution $z_c = 1/\phi$ without imposing ϕ as an input. This axiom carries the full v1.1 lock pattern as algebraic consequences once the v1.1 width relation $w = A/\ln\phi$ is invoked. The axiom can be stated in eight equivalent algebraic forms, each verified to machine precision.

Direct numerical comparison of the frame-matching axiom with the CLS self-duality condition $a(\eta_c) = c_0$ in conformal time, performed on the ΛCDM background, shows that the two conditions identify the same cosmological epoch only at $\Omega_m \approx 0.654$ — in disagreement with Planck [7] at $\sim 50\sigma$ and with the LVC-locked value of $\Omega_m = 0.2907$ at a comparable level. The CLS framework therefore does not contain the LVC frame-matching condition as a special instance on the observational background. The two constructions are structurally analogous but not mathematically equivalent.

The paper is organised in three parts. Part I records the axiom and its algebraic consequences. Part II records the structural-kinship analysis with the CLS framework, including the numerical Planck-disagreement result. Part III records the status of the lock after the total-effort attempt and a closing remark on the structural context.

The discipline of the paper follows that of v1.1, v1.2 and the closure note: phenomenology and reconstruction are kept separate, results are recorded without strengthening claims beyond what the calculations support, and the modal hedge of the closing remark is preserved as part of the result.

Part I — The frame-matching axiom and its algebraic structure

2 Statement of the axiom

In the redshift coordinate z and the scale-factor function $a(z) = 1/(1+z)$, let

$$z_c \;=\; a(z_c) \;=\; \frac{1}{1+z_c}. \tag{1}$$

This is the frame-matching axiom: at the modulation centre, the redshift coordinate equals the scale-factor value at that coordinate.

The condition (1) rearranges to the quadratic

$$z_c(1+z_c) \;=\; 1, \qquad z_c^2 + z_c - 1 = 0, \tag{2}$$

with unique positive solution

$$z_c \;=\; \frac{-1+\sqrt{5}}{2} \;=\; \frac{1}{\phi}, \qquad \phi = \frac{1+\sqrt{5}}{2}. \tag{3}$$

The corresponding values of the scale factor and the logarithmic-redshift coordinate are

$$a_c \;=\; \frac{1}{1+z_c} \;=\; \frac{1}{\phi}, \qquad \theta_c \;=\; \ln(1+z_c) \;=\; \ln\phi. \tag{4}$$

The axiom (1) does not impose ϕ as an input. The golden ratio appears as the unique positive solution of (2), an algebraic consequence. The transcendental identification $\theta_c = \ln\phi$ recorded in v1.1 [1] as an empirical lock is reproduced as the value of $\ln(1+z_c)$ at the algebraic fixed point.

3 Eight equivalent algebraic forms

The axiom (1) admits eight algebraic restatements, each verified to machine precision at $z_c = 1/\phi$:

$$\begin{aligned}
&\text{(P-a)} \quad z_c = a(z_c) && \text{(frame matching)} && (5)\\
&\text{(P-b)} \quad z_c(1+z_c) = 1 && \text{(quadratic form)} && (6)\\
&\text{(P-c)} \quad (1+z_c)^2 = (1+z_c)+1 && \text{(golden recursion)} && (7)\\
&\text{(P-d)} \quad 1/a_c - a_c = 1 && \text{(reciprocal difference)} && (8)\\
&\text{(P-e)} \quad 2\sinh\theta_c = 1 && \text{(hyperbolic half-unit)} && (9)\\
&\text{(P-f)} \quad e^{\theta_c} = 1 + e^{-\theta_c} && \text{(exponential)} && (10)\\
&\text{(P-g)} \quad f(z) = 1/(1+z) \text{ has fixed point } z_c && \text{(map fixed point)} && (11)\\
&\text{(P-h)} \quad \mathrm{gd}(\theta_c) = \arctan(1/2) && \text{(Gudermannian)} && (12)
\end{aligned}$$

Each form is equivalent to the others by algebraic rearrangement, and each recovers $z_c = 1/\phi$ as the unique positive solution.

The hyperbolic form (P-e), $2\sinh\theta_c = 1$, is recorded as the most geometrically transparent: it identifies θ_c as the hyperbolic angle whose hyperbolic sine equals one half. The Gudermannian form (P-h) records the corresponding circular angle as $\arctan(1/2)$, a quantity recurring in number-theoretic contexts.

4 Algebraic consequences of the axiom

The axiom (1) carries the v1.1 lock pattern as algebraic consequences. Given (3) and the v1.1 width relation $w = A/\ln\phi$ [1], the following identities hold by direct substitution and are verified to machine precision:

$$\text{Position lock:} \quad \theta_c = \ln\phi \tag{13}$$

$$\text{Matter–}\Lambda\text{ equality:} \quad \theta_{\mathrm{eq}} = \theta_c \cdot z_c = (\ln\phi)/\phi \tag{14}$$

$$\Omega_m \text{ lock:} \quad \Omega_m = \frac{1}{1+\exp(3\theta_c z_c)} = 0.290653 \tag{15}$$

$$\text{v1.1 amplitude}(k=4): \quad A = (\ln\phi)^4 \tag{16}$$

$$\text{v1.1 width}(k=3): \quad w = (\ln\phi)^3 \tag{17}$$

$$\text{Conformal partner unit:} \quad z_c - z' = 1, \quad a' - a_c = 1, \quad a_c \cdot a' = 1 \tag{18}$$

where z' is the conjugate root $z' = -1/\phi^2$ of (2) and $a' = 1/(1+z')$. The conformal-partner-unit identities follow from the two roots of the quadratic (2) and are recorded for completeness.

The Ω_m lock is the v1.1 recursive relation $\ln\phi = \phi \cdot \theta_{\mathrm{eq}}(\Omega_m)$, restated using $\theta_c z_c = (\ln\phi)\cdot(1/\phi) = \theta_{\mathrm{eq}}$.

5 Status of the closure at the axiom level

The axiom (1) closes the position lock $\theta_c = \ln\phi$ at the algebraic level. The reduction is from the v1.1 lock pattern of four imposed parameters $(\theta_c, A, w, \Omega_m)$ to a single algebraic axiom (frame matching) plus the v1.1 width relation $w = A/\ln\phi$, together with the v1.1 choice of $A = (\ln\phi)^4$.

The axiom is not itself derived from the v1.1 action. It is an algebraic condition between two cosmological variables. The closure is therefore partial in the sense of [3]: closure proceeds by a structural relation internal to the description of the universe, not by inference across instances or by embedding into a larger theory.

The question whether the axiom (1) can in turn be derived from a deeper principle is the subject of Part II, in which the comparison to the CLS scale-factor self-duality framework is performed.

Part II — Structural kinship with CLS self-dual cosmology

6 The CLS self-duality condition

The Camara–Lima–Sotkov scale-factor self-duality framework [4, 5, 6] implements, in flat FLRW cosmology, the conformal-time transformation

$$\tilde{a}(\tilde{\eta}) \;=\; \frac{c_0^2}{a(\eta)}, \qquad \tilde{\eta} \;=\; \pm\,\eta + \text{const}, \tag{19}$$

together with the corresponding transformations of the matter density and pressure, such that the Friedmann equations are invariant in form. The transformation depends on a real constant c_0, which is a free parameter of the framework and represents the scale-factor value at the self-dual fixed point.

The self-duality fixed-point condition,

$$\tilde{a}(\tilde{\eta}_c) \;=\; a(\eta_c) \quad\Longleftrightarrow\quad a^2(\eta_c) \;=\; c_0^2, \tag{20}$$

identifies a specific epoch η_c in conformal time at which the scale factor equals c_0. The choice of the reflection origin ($\tilde{\eta} = -\eta + \text{const}$) determines the location of η_c on the conformal-time axis. Three natural choices are commonly considered in the CLS literature and its extensions:

(I) the midpoint of the conformal-time extent, $\eta_c = (\eta_{\text{past}} + \eta_{\text{future}})/2$;

(II) the present epoch, $\eta_c = \eta_{\text{today}}$, giving $c_0 = a(\text{today}) = 1$;

(IV) the matter–Λ equality epoch, $\eta_c = \eta_{\text{eq}}$.

In each case c_0 is determined by the cosmological background once the convention is fixed.

7 Comparison with the LVC frame-matching axiom

The LVC frame-matching axiom (1) and the CLS self-duality fixed-point condition (20) are both fixed-point conditions on a duality-like map. They are not, however, statements of the same condition. The frame-matching axiom is an algebraic relation between the redshift variable z and the scale-factor function $a(z)$ at a single redshift; it is independent of the cosmological background. The CLS condition identifies the scale factor at a specific conformal-time epoch, the location of which depends on the background through the conformal-time integral.

The question whether, on the standard ΛCDM background, the two conditions identify the same cosmological epoch is decidable numerically.

8 Numerical comparison on the ΛCDM background

Conformal time is computed in the scale-factor variable as

$$\eta(a) \;=\; \int_1^a \frac{da'}{a'^2\,E(a')}, \qquad E(a) \;=\; \sqrt{\Omega_m a^{-3} + (1-\Omega_m)}, \tag{21}$$

in units of $1/H_0$, with $\eta(\text{today}) = 0$, $\eta < 0$ in the past and $\eta > 0$ in the future. The conformal-time extent is finite at both ends: $\eta_{\text{past}} = \eta(a \to 0)$ converges in ΛCDM, and $\eta_{\text{future}} = \eta(a \to \infty)$ converges in any cosmology with $\Omega_\Lambda > 0$.

The LVC frame-matching point sits at $z_c = 1/\phi$, $a_c = 1/\phi$. Its conformal-time location depends weakly on Ω_m and is computed directly from the integral. The CLS midpoint convention (I) places the self-dual point at $\eta_c = (\eta_{\text{past}} + \eta_{\text{future}})/2$, which in turn corresponds to a redshift z_{CLS} depending on Ω_m.

Table 1 records the comparison for three natural values of Ω_m.

Ω_m	$\eta(z = 1/\phi)$	η_{past}	η_{future}	$\eta_c^{(\text{I})}$	$z_{\text{CLS}}^{(\text{I})}$
0.277	−0.5332	−3.4137	+1.1273	−1.1432	1.7537
0.2907	−0.5302	−3.3478	+1.1352	−1.1063	1.6927
0.315	−0.5248	−3.2402	+1.1497	−1.0453	1.5911

Table 1: Conformal-time location of the LVC frame-matching point ($z = 1/\phi$) and of the CLS midpoint self-dual point (convention I), for three natural values of Ω_m (LVC fit, LVC lock, Planck). All quantities in units of $1/H_0$. The two points are separated by a factor of ~ 2 in conformal time at all values tested. The CLS midpoint redshift $z_{\text{CLS}}^{(\text{I})}$ is approximately 1.6, against $z_c = 0.618$ for the LVC point.

The two points are separated, in conformal time, by a factor of approximately two for all Ω_m in the natural range. The corresponding redshifts differ by a factor of approximately 2.6. The CLS midpoint convention does not identify the LVC frame-matching point on the ΛCDM background at any natural value of Ω_m.

9 Inverse search and Planck disagreement

A direct question is whether any value of Ω_m exists for which the CLS midpoint coincides with the LVC frame-matching point. The function

$$\Delta\eta(\Omega_m) \;\equiv\; \eta(z = 1/\phi; \Omega_m) - \tfrac{1}{2}\big[\eta_{\text{past}}(\Omega_m) + \eta_{\text{future}}(\Omega_m)\big] \tag{22}$$

is monotone in Ω_m in the range tested and admits a unique zero at

$$\Omega_m^{\text{match}} \;=\; 0.65380, \tag{23}$$

verified by Brent's method to a residual $|\Delta\eta| < 6 \times 10^{-13}$. At this value of Ω_m the CLS midpoint and the LVC frame-matching point coincide in conformal time, with $c_0 = a_c = 1/\phi$.

The value $\Omega_m^{\text{match}} = 0.654$ is in disagreement with the Planck 2018 result $\Omega_m = 0.315 \pm 0.007$ [7] at the $\sim 50\sigma$ level, and with the LVC-locked value $\Omega_m = 0.2907$ at a comparable level. The CLS midpoint convention does not therefore identify the LVC frame-matching point on the cosmological background consistent with observation.

The alternative conventions (II), (IV) and a half-past convention $\eta_c = \eta_{\text{past}}/2$ were tested for completeness. The corresponding redshifts on the Planck ΛCDM background are $z_{\text{CLS}}^{(\text{II})} = 0$ (trivial), $z_{\text{CLS}}^{(\text{IV})} \approx 0.31$, and $z_{\text{CLS}}^{(\text{VII})} \approx 12.6$. None coincides with $z = 1/\phi$.

10 Status of the kinship

The CLS framework and the LVC frame-matching axiom are recorded as structurally analogous: both are fixed-point conditions on a duality-like map acting on cosmological variables, and both place a scale-factor-related quantity at a privileged value at a specific epoch. The frame-matching axiom is the simpler of the two — it is an algebraic relation independent of the background — and the CLS condition specialises to the LVC point only at a non-physical value of Ω_m.

The relation between the two is therefore at the level of *structural kinship*, not *strict embedding*. The LVC frame-matching condition is not a special case of the CLS self-duality framework on the observational background. The CLS framework does not, on the analysis recorded here, contain the value $c_0 = 1/\phi$ as a forced instance: the framework leaves c_0 free, and the cosmologically natural conventions yield values of c_0 distinct from $1/\phi$.

This result places a structural constraint on attempts to derive $\theta_c = \ln\phi$ from the CLS framework: such a derivation would require either (a) an additional principle, internal to CLS, that selects $c_0 = 1/\phi$ from among the free-parameter values, or (b) a modified self-duality construction in which the fixed-point condition is replaced by the frame-matching condition. Neither route is pursued in the present paper.

Part III — Status and closing remark

11 Status of the lock after total-effort closure

The empirically preferred modulation centre $\theta_c = \ln\phi$ of the LVC construction has been examined under a sustained closure attempt summarised as follows.

1. The position is closed at the algebraic level by the frame-matching axiom $z = a(z)$ at the modulation centre. The axiom carries the v1.1 lock pattern $(\theta_c, A, w, \Omega_m)$ as algebraic consequences once the v1.1 width relation $w = A/\ln\phi$ and the v1.1 choice $A = (\ln\phi)^4$ are invoked. Eight equivalent forms of the axiom are recorded in Part I.
2. The axiom is not itself derived from the v1.1 action or from any deeper principle. The reduction is from four imposed parameters to one algebraic condition; the condition itself remains a posited structural relation.
3. The CLS scale-factor self-duality framework is structurally analogous to the frame-matching axiom but does not contain it as a special instance on the cosmological background consistent with observation. The CLS midpoint convention would identify the LVC point at $\Omega_m = 0.654$, in $\sim 50\sigma$ tension with Planck. The two frameworks are siblings at the level of fixed-point construction on a duality-like map; they are not in an embedding relation on the ΛCDM background.
4. Across a survey of more than thirty distinct candidate constructions (cosmological-internal, mathematical, quantum-geometric, and topological), no closed-universe forcing of the value $\theta_c = \ln\phi$ from a single non-circular principle has been found. The pattern of resistance is uniform: each candidate either reproduces a near-neighbour value of θ_c or requires inputs equivalent to imposing the answer.

The status of the lock is therefore: closed at the level of a single algebraic axiom internal to the closed-universe theory; not closed by embedding into the CLS framework; and not closed in the strong sense of a forcing derivation from a deeper non-circular principle.

This status is recorded as the outcome of the closure effort. The categorical analysis of [3] provides the structural account of why such a strong closure is not expected to be available within any cosmological theory under the encompassing definition. The frame-matching axiom is the maximal closure achieved within the closed-universe setting in the present series.

12 Closing remark

The persistent resistance of $\theta_c = \ln \phi$ to complete internal derivation in the closed-universe setting may itself be structurally unsurprising. In many natural systems, appearances of the golden ratio are not imposed axiomatically but emerge from optimisation, stability, or resonance-avoidance processes. If cosmological parameters belong to the same category of emergent structural quantities, the absence of a uniquely forcing derivation may reflect not merely incompleteness of the model, but the characteristic manner in which such constants arise in nature.

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